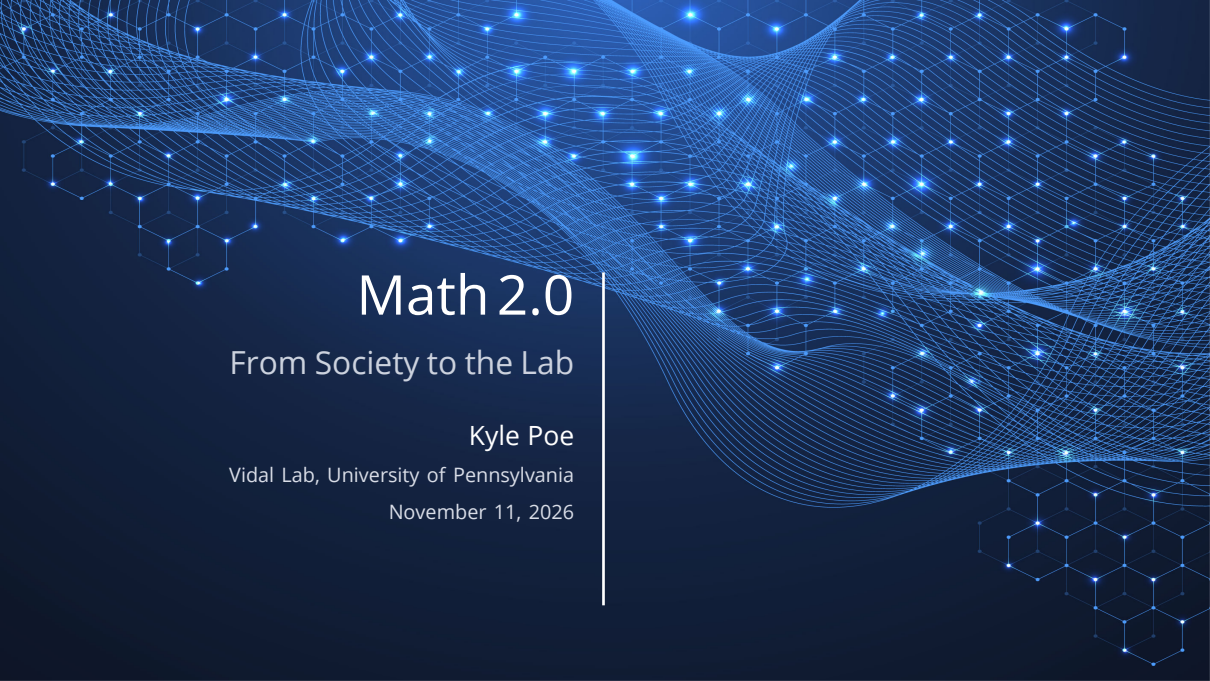


Math 2.0

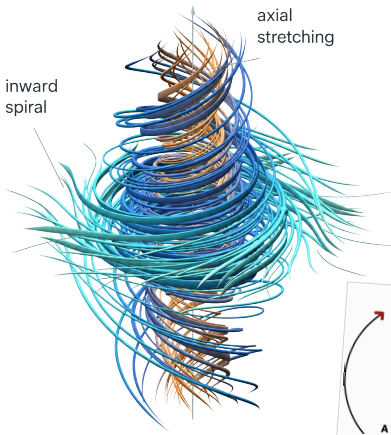
From Society to the Lab

Kyle Poe

Vidal Lab, University of Pennsylvania

November 11, 2026

September 8, 2026: Math changes forever



OpenAI @OpenAI

We're sharing a solution to the Navier-Stokes Millennium Prize Problem, one of the deepest problems at the frontier of mathematics.

nature

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NEWS | 08 September 2026

OpenAI claims huge maths
gh on a famed
n Problem'

cked the physics of fluids
in mathematics.

OpenAI Claims to Have Solved a $1M Math Problem: A Credit Dispute Has Already Begun

Ten thousand AI agents resolved one of math's oldest open problems in 88 hours. A credit dispute over authorship erupted almost immediately.

BY TECHNOLOGY DESK · SEPTEMBER 9, 2026 · 3 MINS READ

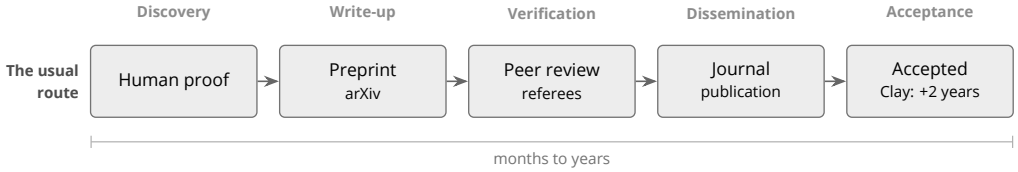


OpenAI Cracks 200-Year-Old 'Millennium' Math Problem—But Credit Dispute Looms

An NYU math professor claims the AI behemoth rushed to solve the problem after discovering his progress.

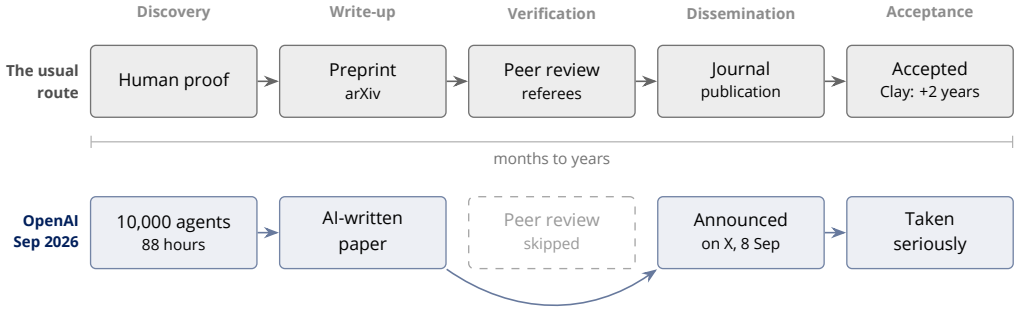
Sources: OpenAI on X, 8 Sep 2026 • Nature news, 8 Sep 2026 • Technology Desk, 9 Sep 2026 • Forbes • Visualization: OpenAI • Meme: @MozeTech on X

Why believe it?



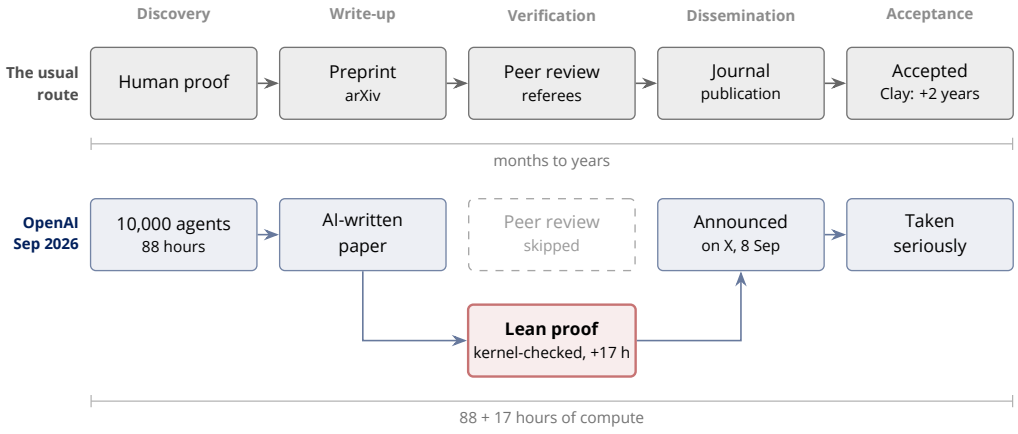
Sources: OpenAI, 8 Sep 2026 • Reisinger et al., *The Mathematical Sciences in an AI-enabled World*, AcadMathSci, Oct 2026 • Clay Mathematics Institute, Millennium Prize rules

Why believe it?



Sources: OpenAI, 8 Sep 2026 • Reisinger et al., *The Mathematical Sciences in an AI-enabled World*, AcadMathSci, Oct 2026 • Clay Mathematics Institute, Millennium Prize rules

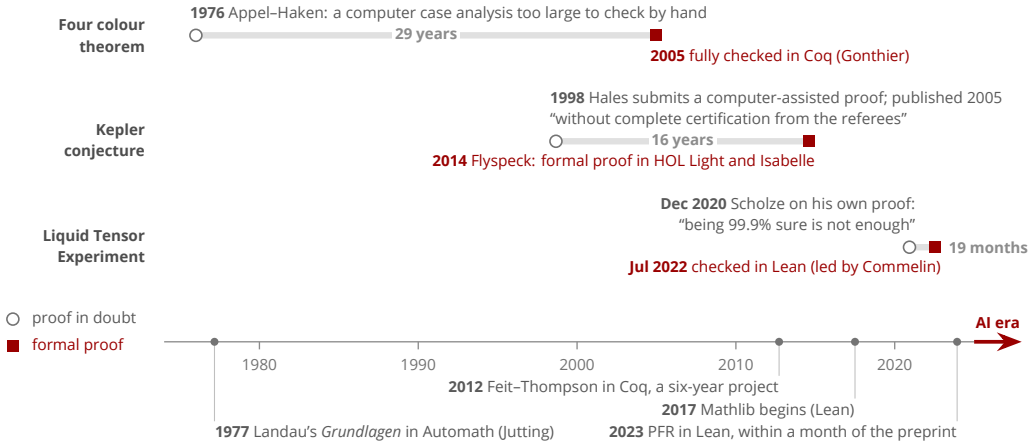
Why believe it?



Lean verification, not merely OpenAI's reputation, made the community take this seriously.

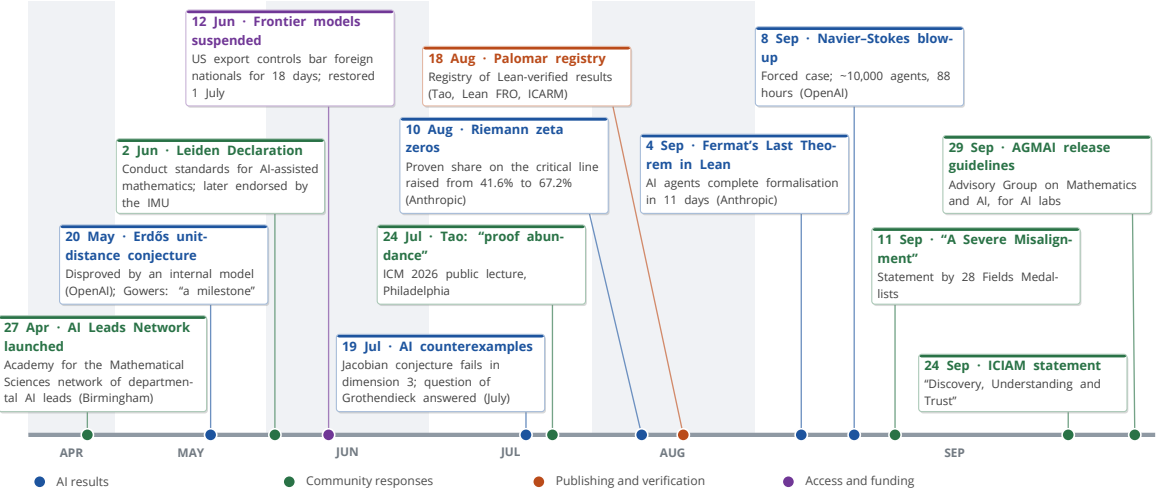
Sources: OpenAI, 8 Sep 2026 • Reisinger et al., *The Mathematical Sciences in an AI-enabled World*, AcadMathSci, Oct 2026 • Clay Mathematics Institute, Millennium Prize rules

Timeline: Formalization in the Pre-AI Era



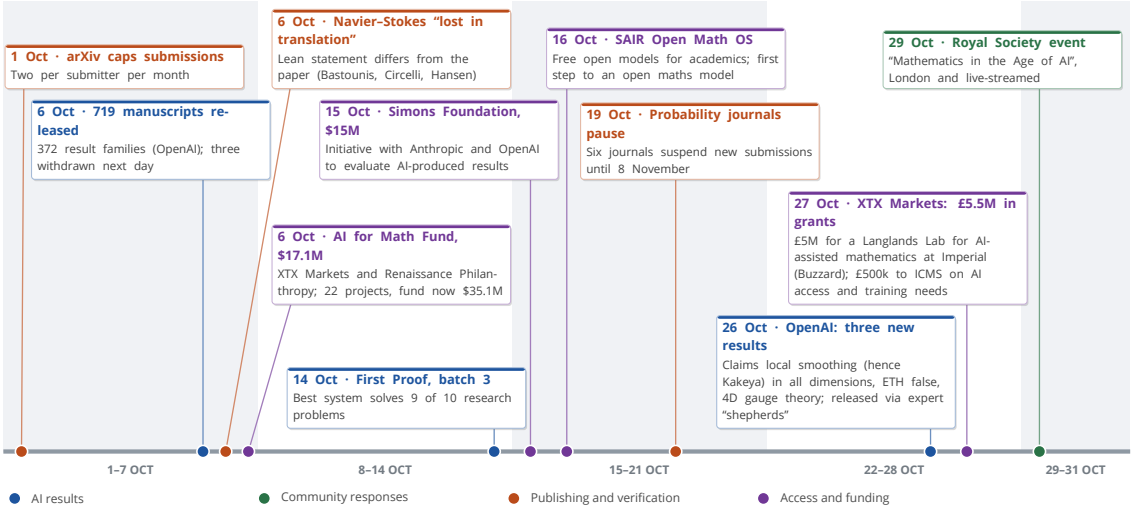
Sources: Jutting, thesis (1977) • Gonthier, MSR report (2005) • Hales, Annals 162 (2005); Hales et al., arXiv:1501.02155; Flyspeck repo • Gonthier et al., ITP 2013 • Scholze, Xena blog (Dec 2020); Lean blog (Jul 2022) • mathlib Community, CPP 2020 • Gowers et al., arXiv:2311.05762; Tao, blog (Dec 2023)

Timeline: AI Transforms the Mathematical Sciences



Adapted from Reisinger et al., *The Mathematical Sciences in an AI-enabled World*, AcadMathSci, pre-publication draft, Oct 2026, Fig. 1 (April–September 2026)

Timeline: AI Transforms the Mathematical Sciences



Adapted from Reisinger et al., *The Mathematical Sciences in an AI-enabled World*, AcadMathSci, pre-publication draft, Oct 2026, Fig. 1 (October 2026)

Math 1.0 and Math 2.0

	Light AI use	Heavy AI use
Weak human understanding	**Pre-math**	**Misaligned math**
Strong human understanding	**Math 1.0**	**Math 2.0**



> *[Rather than blindly using AI to generate misaligned mathematics,] the more sustainable* "Math 2.0" *use of AI would be to strengthen all the ways in which human understanding interacts with mathematics.*
>
> — Terence Tao, *Machine Assisted Proof*, AIM Public Lecture, Caltech, 9 Oct 2026

AI will keep making proofs cheaper. Making formalization a standard step in research is how that abundance raises both scientific integrity and individual productivity, instead of trading one for the other.

Quote and diagram: T. Tao, *Machine Assisted Proof*, AIM Public Lecture Series, Caltech, 9 Oct 2026 (caltech.edu/campus-life-events/calendar/aim-public-lecture-series-terence-tao-1)

The rest of this talk

PART I
Society

- Checking did not scale
- Provenance is unknowable
- Certificates

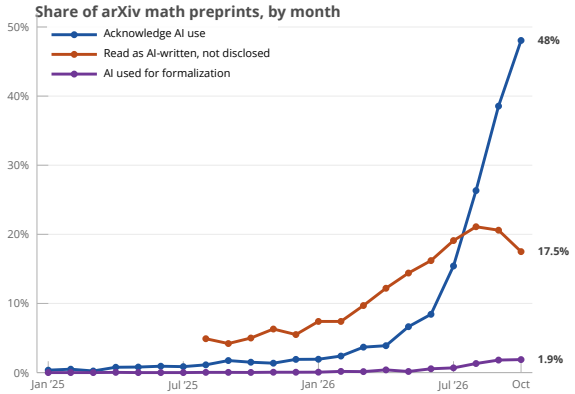
PART II
The Lab

- Learning theory
- The workflow
- In the lab

Part I

Society

The Poison Fountain



1. **Use is exploding.** Disclosed AI use went from 0.4% to 48% of math preprints in 21 months, doubling about every three months. Math submissions are up 75% on a year earlier (Sep 2026).

2. **Much of it is undisclosed.** About 1 in 6 math preprints read as AI-written without saying so. At ICLR 2026, 9% of submissions were mostly AI-written.

3. **Few certify.** OpenAI shipped Lean proofs with 42% of its 719 manuscripts. Across arXiv math, 1.9% acknowledge AI for formalization; in stat.ML, 0.4% even mention Lean.

4. **Review is overwhelmed.** 21% of ICLR 2026 reviews were fully AI-generated. arXiv capped submissions (1 Oct); six probability journals paused (19 Oct – 8 Nov).

Epoch AI (CC BY 4.0, 9 Nov 2026) • K. Martin (3 Nov 2026) • our arXiv count • Pangram, ICLR 2026 (self-reported) • AcadMathSci, Oct 2026

Checking did not scale

Where	Evidence
Volume	arXiv math: 7,428 submissions in Sep 2026, against 4,252 a year earlier (+75%). OpenAI released 719 manuscripts in a single day (6 Oct)
Referees	Six probability journals paused new submissions (19 Oct – 8 Nov). Editors report they cannot find a single referee; nobody counts the declines
Formal libraries	Mathlib: 65 reviewers for 950 contributors (reported)
AI reviewing	ICLR 2026: 21% of reviews fully AI-generated
Bypassed review	Results announced on X and GitHub before any review: Navier–Stokes (8 Sep), the 719 manuscripts (6 Oct)

Production scales with compute. Expert checking scales with experts.

AcadMathSci, Oct 2026 (§1.2, §4.7) • Nee, Zhong & Ni, arXiv 2609.04872 • Weinreich, arXiv 2608.02859 • Pangram, ICLR 2026 (self-reported) • OpenAI, 8 Sep and 6 Oct 2026

Who wrote it?

17.5%

of math preprints read as AI-written without saying so (Oct 2026).

Likely an undercount: the detector flags only half of the papers that *do* disclose. (Martin)

22.5%

of ICML 2026 reviewers used an LLM under a no-LLM policy.

Randomized trial of review policies. (Kim et al.)

No effect

AI-assisted writing grew at the same rate in journals with and without an AI policy.

5,114 journals. (arXiv 2512.06705, via AcadMathSci)

A check that depends on knowing who wrote the proof will fail.

The check has to read the proof, not the author.

K. Martin, AI use in arXiv mathematics papers (3 Nov 2026) • Kim et al., arXiv 2609.19420 • arXiv 2512.06705, cited in AcadMathSci, Oct 2026 (§1.1)

What institutions tried

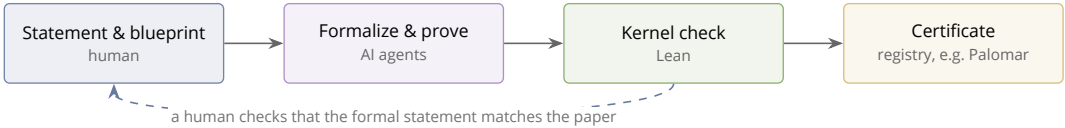
Strategy	Examples	Does it hold?	Status
Restrict	arXiv: two new submissions per submitter per month (1 Oct). Six probability journals paused. *Phil. & Public Affairs* bans AI-written papers	Caps are a stopgap; the pauses ended 8 Nov. Bans depend on detection: "welcome but unenforceable"	• partial
Disclose	Leiden Declaration (endorsed by the IMU); mandatory AI statements (Springer); SIAM AI policy v2.0	Widely ignored (ICML: 22.5%); no measurable effect on AI writing	• partial
Expand	Paid, AI-assisted refereeing (Maltsev); credit for refereeing (AcadMathSci)	Needs funding, and no venue yet trusts AI review with the deciding vote	• open
Verify	Palomar, a registry of Lean-verified results (18 Aug). A proposed journal requiring Lean proofs (Guo, 17 Oct). AcadMathSci: assess correctness "utilising AI and formalisation"	Scales with production and does not depend on who wrote the proof. Still prototypes	• open

Only verification scales with production, and it is the least built.

AcadMathSci, Oct 2026 (§1.2, §4.6, C.1) • Times Higher Education, May 2026 • Kim et al., arXiv 2609.19420 • Maltsev, Proofs & Prompts • Tao, blog, 18 Aug and 17 Oct 2026

AI-assisted formalization

AI translates mathematics into a proof assistant such as Lean and fills in the proofs. A small trusted kernel checks every step. Humans write, or review, what is being proved.



What it addresses

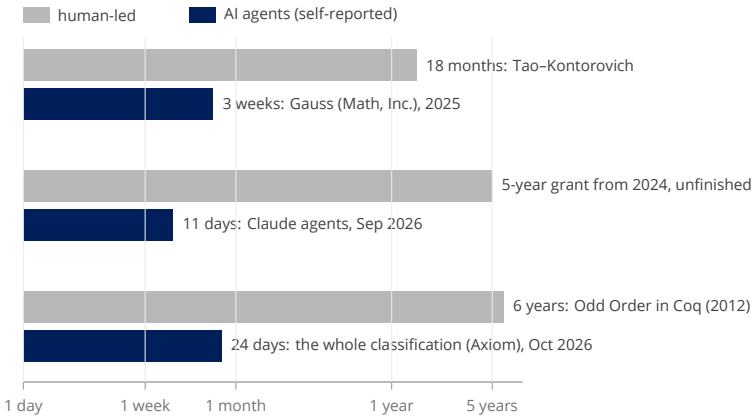
- **Review capacity.** Referees review a page of statements, not 60 pages of proof.
- **Undisclosed AI.** The check does not depend on who, or what, wrote the proof.
- **Bypassed review.** Verify and register a result before announcing it.
- **Credit.** Registries record what produced each result.

What stays open

- **Statement fidelity.** Compiling is not saying the same thing (Navier–Stokes).
- **Understanding.** Millions of machine-written lines explain nothing.
- **Significance and ideas.** Neither is a formal property.
- **Access and coverage.** Frontier compute; thin libraries in PDE and applied probability.

Tao, *Palomar*, blog, 18 Aug 2026 • Bastounis, Circelli & Hansen, arXiv 2610.08144 • research notes 01–03

Formalization got cheap



Log scale. The AI-era figures come from the companies; the finite-simple-groups build has not been independently reproduced.

Math, Inc., Strong PNT (2025) • Anthropic, FLT (Sep 2026) • Axiom, CFSG (Oct 2026) • Gonthier et al., ITP 2013 • research note 01, §1

Certificates catch what review missed

OpenAI's 719 manuscripts

Sep–Oct 2026 • independent audit

- 3 withdrawn within days after a sign error. **All three were outside the Lean coverage.**
- An audit of the 245 formalized families found no theorem shown false.
- But 21 headline results labelled "(Lean)" had nothing formalized: check the certificate, not the label.

Finite simple groups

Axiom, Oct 2026 • self-reported, build not yet reproduced

- 34 deviations from a published record that experts had checked for 40 years.
- A false equality in a published quasithin lemma: a counterexample in $SL_2(5)$.
- An unjustified step in the Unbalanced Group Theorem (1986), repaired by a new route.

The Stacks Project

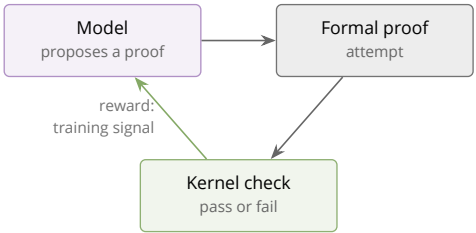
Axiom, Nov 2026 • self-reported

- Dozens of statements that fail in the stated generality: missing hypotheses, edge cases in positive characteristic.
- Each Lean declaration links to its Stacks tag, so a human can check one statement at a time.

Before AI, formal proofs settled doubts one theorem at a time. Now they find errors at scale.

borism/openai-math-slop, issue #2 • OpenAI release record (Wikipedia) • Axiom, CFSG (Oct 2026) and Stacks (Nov 2026) • research note 01, §§2–3, 5

No verifier, no flywheel



AlphaProof autoformalized about 1M problems into about 80M Lean statements to train on this loop.

- Where a result can be checked automatically, every attempt is a training signal. AcadMathSci: Lean is becoming the way AI checks its own work.
- Where it can't (modelling choices, significance, exposition), a person has to judge every output.
- So capability "will stay uneven across the mathematical sciences".

AI gains arrive first where results can be checked. Formalizing a field puts it on that side.

Reisinger et al., AcadMathSci, Oct 2026 (§1.2; A.5, the report's assessment)

The paper and the code disagree

Navier–Stokes (Bastounis, Circelli & Hansen, Oct 2026)

- The paper's estimate (8.19) needs $m+4$ derivatives; the matching Lean theorem assumes $m+5$, and the pattern repeats.
- The paper's pressure bound depends on one quantity; the Lean bound depends on two, and is proved by a different route.
- They do not claim the paper is wrong.

OpenAI's 719 manuscripts (community audit of 245 formalized families)

- About 210 consistent at the level of statements; no theorem shown false.
- 21 headline results marked "(Lean)" were not formalized; 5 formalized only a supporting lemma.

> *Lean acceptance (compilation) of translation does not imply semantic preservation.*
>
> — Bastounis, Circelli & Hansen, arXiv 2610.08144

The drift was found because there was formal code to hold the paper against.

A certificate proves *a* theorem. Whether it is *the* theorem is a human question.

Bastounis, Circelli & Hansen, arXiv 2610.08144 • borism/openai-math-slop, issue #2 • research note 01, §5

Three layers of checking

1. Kernel

Does Lean accept the proof? Cheap and mechanical. Watch for `sorry`, extra axioms and kernel bugs. • in place

2. Statement

Does the formal statement say what the paper claims? Scarce: "will never be solved purely in software" (Kontorovich). Navier–Stokes drifted here. • open

3. Understanding

Does anyone understand the proof, and does it match the written one? Scarce. Machine proofs run to millions of lines. • open

Layer 1 makes correctness cheap and auditable. Layers 2 and 3 stay human: Part II.

Research note 01, §0 • Kontorovich, via AcadMathSci, Oct 2026 (§2.2) • Bastounis, Circelli & Hansen, arXiv 2610.08144

Part II

The Lab

Formalization reaches learning theory

[1.5 min] It is starting, and there is room for us: StatsMLlib, TorchLean, the stat.ML count, a Lean-proved learning-theory paper. Needs Q15/Q17 research

Why learning theory needs statement discipline

[1 min] Our theorems are inequalities with constants and assumptions: exactly where Navier–Stokes drifted (note 05, s6). Certificates and statement review together

Running example

[1.5 min] A lemma from the lab's own work, carried through the four stages that follow. Also the best evidence for the productivity claim

State: you own the statement

[2.5 min] L2. Write and review statements yourself; blueprint the proof. Tools: leanblueprint, Verso Blueprint, Comparator challenge files. Navier–Stokes drift as the warning

Prove: let the machine fill it in

[2 min] L1. Delegate the proof, not the statement. Tools: LeanCopilot, lean-lsp-mcp, Aristotle, AXLE, Gauss; libraries: Mathlib, StatsMLlib

Check: read what compiled

[2 min] L1. #print axioms; no sorry or native_decide; pinned toolchain; re-check with –trust=0 after soundness advisories (the Collatz exploit, built deliberately by Ramana Kumar)

Understand and share

[2 min] L3. Hand-picked lemmas as the human-facing narrative over a machine-built proof; register and disclose (Palomar, formalization.yaml)

A spectrum of human direction

[1 min] One slide, one row per case: fully automatic (Navier–Stokes), agents on a human roadmap (FLT), AI filling a blueprint (sphere packing), human-led (Armstrong–Kempe)

Beyond proofs: code and experiments

[1.5 min] The same principle as Part I: certify the output, not the process. Tests, validated numerics and documented validation are the certificate for code and experiments (AcadMathSci s4.6; Haugen–Stepanenko–Hansen)

Budget and access

[1.5 min] AcadMathSci App. B costs; plan for top-tier prices; access gap (11% funded; SAIR Open Math OS). Absorbs the access row moved out of Part I

Checklist

[1 min] One item per workflow stage: state, prove, check, understand and share

Closing

Where this leaves us

[0.5 min] Recap the thesis: formalization as a standard step buys integrity (Society) and productivity (The Lab)

Proof abundance, insight scarcity

[1 min] Full circle with the opener and Tao: certificates make correctness cheap, which frees attention for statements and understanding; insight is what stays scarce, and it lives in individuals

Stay the author

[1 min] Individual responsibility: keep control of what is generated, keep your own insights, do not cede your perspective to AI. Anchors moved here from Part I: deskilling (Bastani, -17%), narrowing (Hao), the convex hull

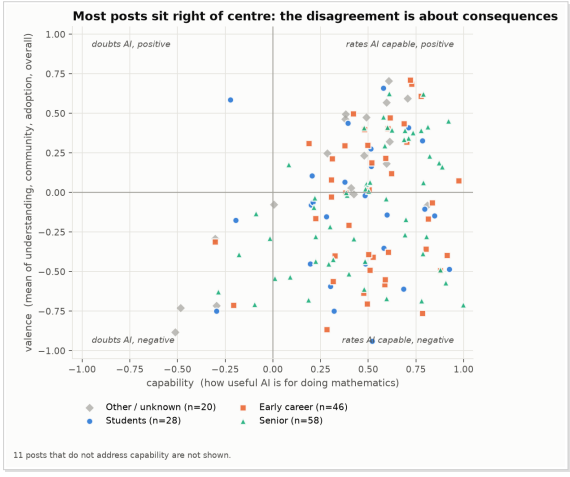
A call to action

[0.5 min] A few personal commitments for each person in the room

Discussion

[– min] Prompts to open the 15 minutes

Proofs and Prompts: A Sentiment Analysis



152 essays on *proofsandprompts.com*, each scored by an LLM

89%

rate AI as capable of doing mathematics

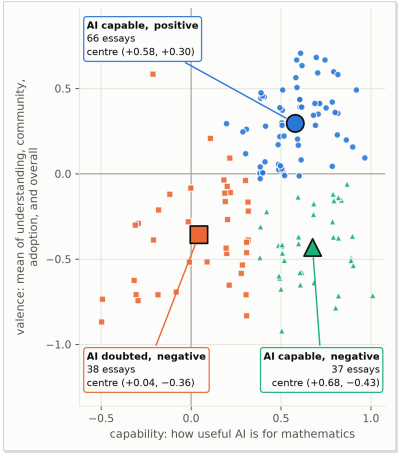
49%

view AI as a negative force overall

The field agrees that it works.
It disagrees about the consequences.

Proofs & Prompts corpus (proofsandprompts.com), scored with Claude; analysis set of 152 posts, 141 addressing capability. Self-selected essays: an illustration, not a survey

Proofs and Prompts: What Each Cluster Writes About





Proofs & Prompts corpus; scores from Claude (rubric.py); clustering and word statistics from cluster.py. Self-selected essays: an illustration, not a survey

Credit, disclosure and confidentiality

Problem	Case	Response	Status
Credit	AI reuses known ideas without attribution: 8 of Aletheia's 13 meaningful Erdős results were already in the literature. Kolda: "Human–AI scrapple"	Leiden Declaration (Jun 2026, endorsed by the IMU): human responsibility, attribution	• partial
Priority	Navier–Stokes priority dispute (Córdoba & Martínez-Zoroa; Buckmaster & Alpöge)	AGMAI release guidelines (29 Sep): review before announcement, release prompts. Labs withheld prompts	• open
Disclosure	1 in 6 math preprints read as AI-written without saying so. At ICML 2026, 22.5% of reviewers used an LLM under a ban	Disclosure statements (arXiv, Springer, Leiden); SIAM AI policy v2.0	• partial
Confidentiality	Referees uploading manuscripts to AI tools	LMS: misconduct. SIAM v2.0: "non-delegation of scholarly judgement"	• partial

The rules already bind individuals; none of them can be checked from the paper alone.

Reisinger et al., AcadMathSci, Oct 2026 (A.3, C.1, C.4, C.5) • Kolda, Silicon Reckoner, Feb 2026 • Martin, Nov 2026 • Kim et al., arXiv 2609.19420

Training, access and narrowing

Problem	Case	Response	Status
Training	Students given unrestricted GPT-4 scored 17% lower once access was removed; a tutor with guardrails largely avoided the harm (Bastani et al.)	Vivas and in-person exams; AI-permitted work graded on formulation and validation	• partial
Access	Only 11% of researchers have institutional funding for their main AI tool (Springer Nature, 10,480 researchers). Navier–Stokes cost an estimated £4.9M	Token costs eligible on grants (proposed); SAIR Open Math OS: free open models for academics	• partial
Narrowing	AI adoption shrinks topic range by 4.6% and engagement by 22% (Hao et al., 41M papers). Agendas drift toward "problems easily amenable to current AI"	Reward understanding and long-term contribution over volume (AcadMathSci)	• open

Reisinger et al., AcadMathSci, Oct 2026 (§1.6, §3.2, §3.3, §4.3, B.1) • Bastani et al., PNAS • Hao et al., arXiv 2412.07727

Corpus and scoring method

[– min] Details behind the opinion map

Backup slides

[– min] Material for questions

References

[– min] Sources